(Please write your Enrollment Number) Enrollment No. 194

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MID-TERM EXAMINATION
Programme: B.Tech (CSE-AI) (Semester: VI)
(March 2026) OFF LINE mode

Subject Code: BAI 306	Subject: Digital Image Processing
Time : 1 ½Hours	Maximum Marks :30

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

Q1		
	a) What is image sampling and quantization? How do these processes affect the quality of a digital image?	(2.5*4=10)
	b) Explain with examples of contrast stretching and thresholding gray-level transformations.	
	c) Differentiate between spatial domain filtering and frequency domain filtering.	
	d) Explain ringing effect. Why it is obviously seen in the Ideal Lowpass Filter results?	
	UNIT I	
Q2	**Attempt any two parts**	
	a) Describe the fundamental steps in a typical digital image processing system. How does each step contribute to the final output?	(5*2=10)
	b) Consider the following 5 × 5 binary image (0 = background, 1 = foreground object): $\begin{matrix} 0 & 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{matrix}$ Let the set of pixels be V={1}, and let the center pixel be P=(2,2). Answer the following: 1. Give the 4-neighborhood and 8-neighborhood of P. 2. Which pixels are m-adjacent to P? 3. Using 4-connectivity how many connected components are present in the image? 4. Calculate the D_4 distance from P(2,2) to A=(4,4). Calculate the Dm distance from P(2,2) to A=(4,4).	
	c) Perform Histogram Equalization Rk: 0 1 2 3 4 5 6 7 nk: 25 41 64 62 92 110 169 221 Draw histograms for the input and output images.	
	UNIT II	
Q3	**Attempt any two parts**	
	a) Consider the following 5 × 5 grayscale image: $f(x,y) = \begin{matrix} 1 & 2 & 3 & 2 & 1 \\ 2 & 4 & 6 & 4 & 2 \\ 3 & 6 & 9 & 6 & 3 \\ 2 & 4 & 6 & 4 & 2 \\ 1 & 2 & 3 & 2 & 1 \end{matrix}$ Apply the following filters only at the center pixel C (2,2). 1. Apply 3×3 Mean Filter 2. Apply 3×3 Weighted (Gaussian) Filter 3. Apply a median filter 4. Apply a simple Laplacian Filter Apply the Sobel Operator to compute Gx, Gy, and the gradient magnitude |G|	(5*2=10)
	b) Explain the concepts of Ideal Low-Pass Filter (ILPF), Butterworth Low-Pass Filter (BLPF), and Gaussian Low-Pass Filter (GLPF) in the frequency domain.	
	c) Explain the significance of the Fourier Transform in digital image processing. How does converting an image to the frequency domain help in filtering operations?	

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